

MEDICARE MONTHLY ENROLLMENT DASHBOARD

Digital Service at Centers for Medicare & Medicaid | Open Source Program Office

Kiran Sheikh	Software Engineering Fellow <i>UC Irvine '25, Computer Science and Business Information Systems</i>
Haseeb Malik	Software Engineering Fellow <i>Brooklyn College, Computer Science</i>
Isaac Guzman	Software Engineering Fellow <i>Columbia University, Computer Science and Film Media</i>
Zion Muhammad	Software Engineering Fellow <i>University of Michigan, Computer Science</i>
Julio Jijon	Software Engineering Fellow <i>Berea College, Computer Science</i>

Keywords:

Full-Stack, API Endpoints, Data Visualizations, Open Source (Practices and Standards), JavaScript, UI/UX, Mobile Development

Summary:

To make Medicare monthly enrollment data accessible and explorable by the public, the CIF fellows implemented an Open Source dashboard that pulls directly from CMS's public dataset API and renders enrollment metrics as interactive maps, trend charts, and data grids.

Statically developed with **Eleventy**, **D3.js**, and the **US Web Design System**, the site calculates the latest monthly figures into a modern client-side interface, accessible on both desktop and mobile web browsers.

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coding it forward >



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Coding it Forward Team (Summer '26)

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Software Engineering Fellows



Kiran Sheikh

*UC Irvine '25,
Computer Science
and Business
Information Systems*



Haseeb Malik

*Brooklyn College,
Computer Science*



Isaac Guzman

*Columbia University,
Computer Science
and Film Media*



Zion Muhammad

*University of
Michigan,
Computer Science
(Master's)*



Julio Jijon

*Berea College,
Computer Science*

What are the Centers for Medicare and Medicaid Services? (CMS)

“CMS is the United States federal agency that provides health coverage to more than **160 million** people through **Medicare, Medicaid, the Children's Health Insurance Program, and the Health Insurance Marketplace.**”



70,330,194

people enrolled in Medicare as of April 2026

What is an Open Source Program Office? (OSPO)

An OSPO is responsible for defining and implementing programs, policies, and projects that guide organizational open source efforts.

30% of Fortune 100 companies have implemented an OSPO, and CMS is the first United States federal agency to implement this private-sector styled approach to an Open Source Strategy.

The summer '26 CIF team worked alongside these dedicated civil servants to help solve some of the most complex problems facing the healthcare system today.



Save us Money



Save us Time



Reduce Duplicate Work



Reduce Duplicate Costs



Reduce Security Risk



Reduce Continuity Risk

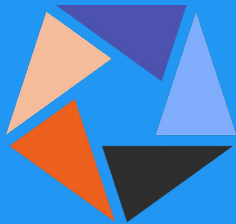


Accountability for Contract Performance



Engine for Talent

MEDICARE MONTHLY ENROLLMENT DASHBOARD OVERVIEW



PROJECT SCOPE

- Background** CMS publishes a dashboard of the latest enrollment counts of Medicare beneficiaries, visualizing hospital/medical and prescription drug coverage by geographic area.
- Problem** The dashboard is generated by a proprietary tool, has no mobile support, and does not meet accessibility standards.
- Our Goal** A full re-implementation of the Medicare Monthly Enrollment Dashboard (modern features, better performance, and a static open-source stack).

Rebuild in open source

Modernize the UI

Deploy as a static site

ORIGINAL DASHBOARD

Original dashboard (proprietary tool, not Open Source)

Medicare Enrollment for April 2026

Hospital/Medical

Prescription Drug

MA: 36.0M
(51% of total)

70.3M
Total
Enrollment

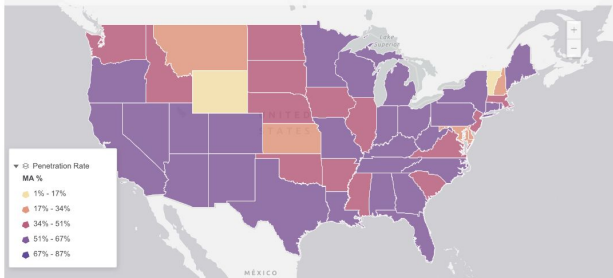
FFS: 34.3M
(49% of total)

Map/Graphs

Grid

All Areas Counties

Medicare Advantage (MA) & Other Health Plans Penetration Rate



Esri, TomTom, Garmin, FAO, NOAA, USGS, EPA, USFWS

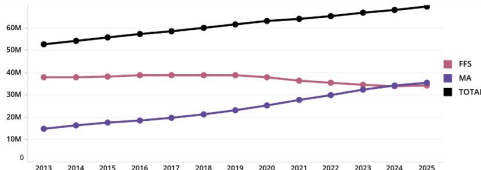
Powered by Esri

*Map includes data for all 50 United States plus Washington D.C. Click to view state level information. See "Grid" for additional US territories.

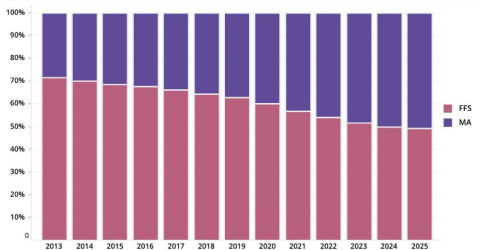
Yearly Trend

12-Month Trend

Enrollment Count Yearly Trend: All Areas



Percent of Total Enrollment Yearly Trend: All Areas



Statistics provided from Lighthouse on Google Chrome developer

40

92

Performance Accessibility

Values are estimated and may vary. The [performance score is calculated](#) directly from these metrics. [See calculator.](#)

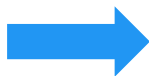
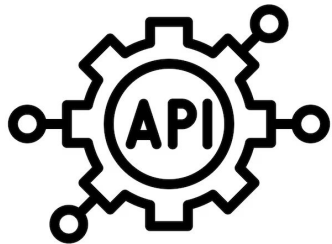
▲ 0-49

■ 50-89

● 90-100

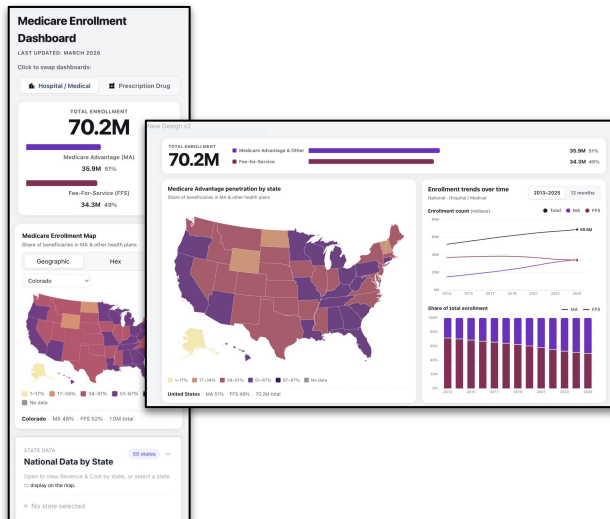
ITERATIVE PROCESS

API Fetch Configuration
(Backend Layer)

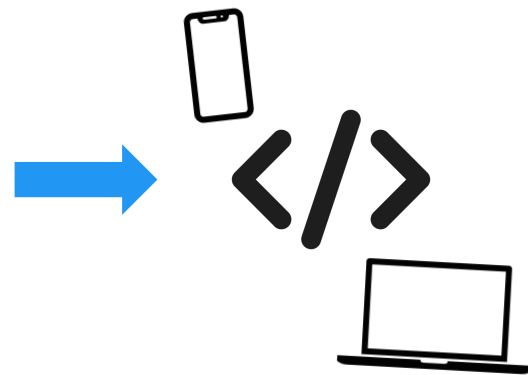


Mockup Design

Stakeholder Feedback



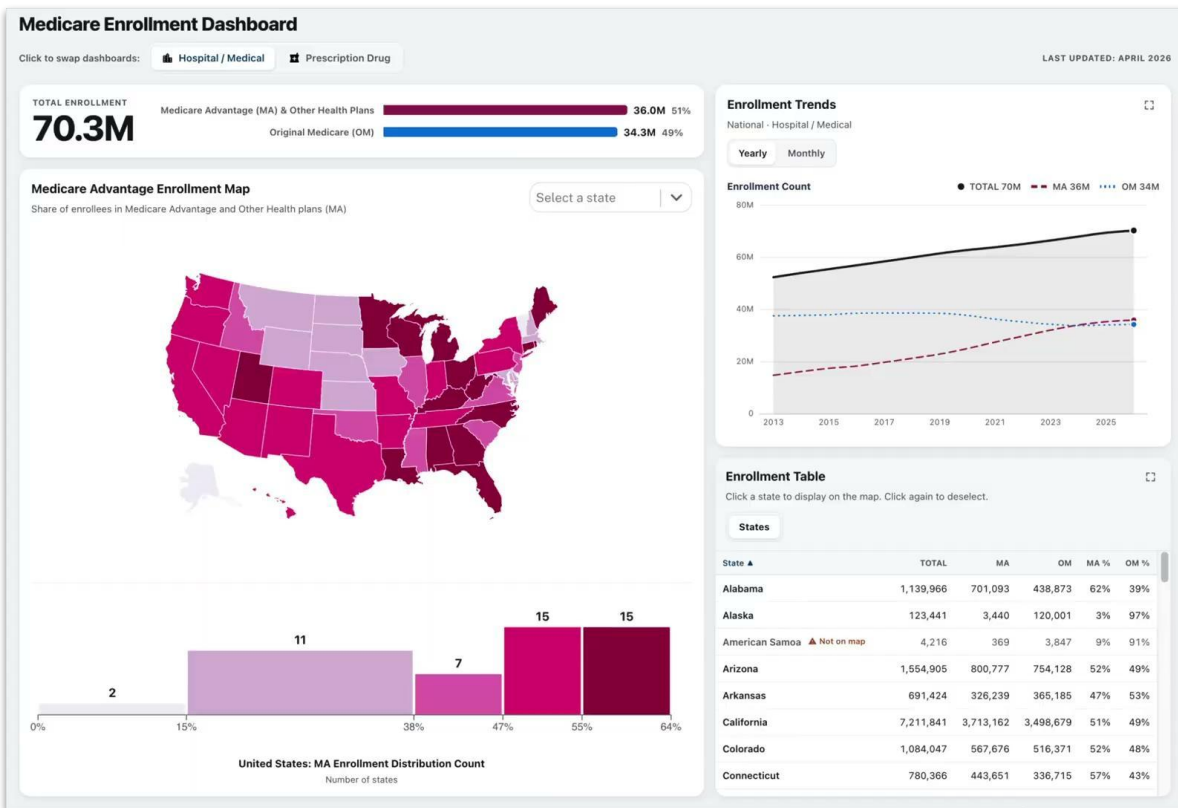
Frontend Mobile &
Desktop Implementation



DESKTOP BUILD

Card Layout

1. **Hero Card** - total metrics
2. **Map Card** - interactive, select state and county
3. **Trend Card** - data over time (line chart, bar chart, table)
4. **Grid Card** - interactive, data by state/county (tables)



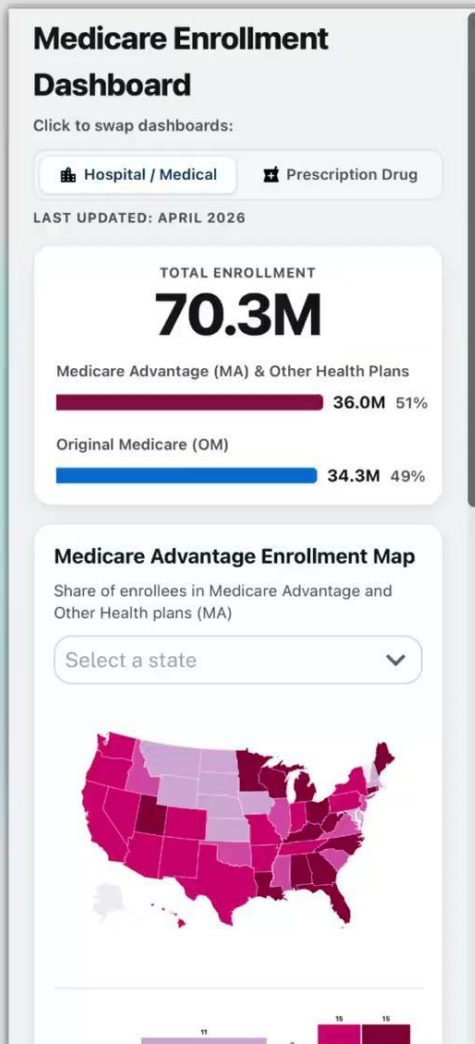
MOBILE BUILD

- **Concept**

- All the same “cards” as desktop view
- Turns into Mobile view based on a trigger state
= crossing screen-width threshold

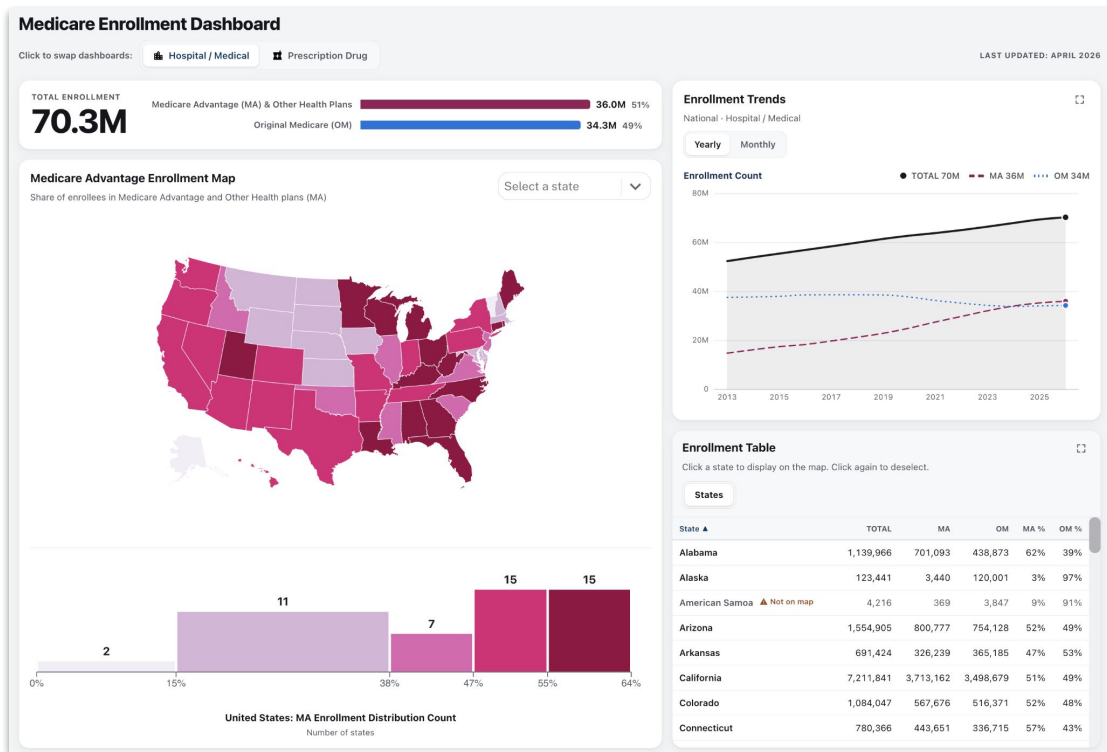
- **Mobile-specific features**

- Trend card (holds data visuals) turns into a widget
= simple swipe between visuals
- Data tables turn into a mobile drawer pop-up
= does not interrupt vertical scroll

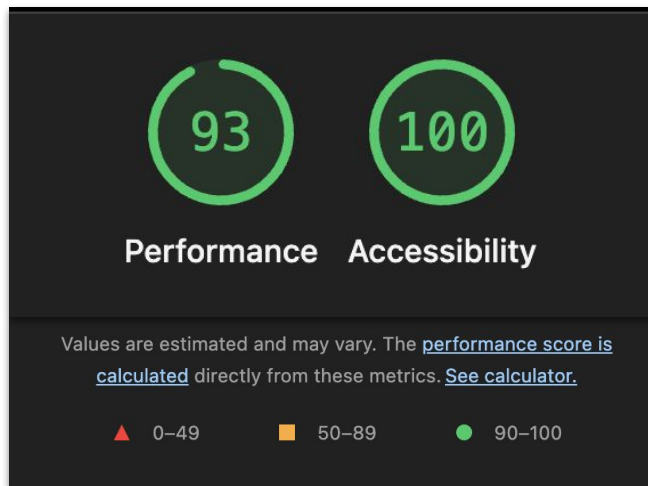


NEW DASHBOARD

New dashboard (Live and Open Source!)



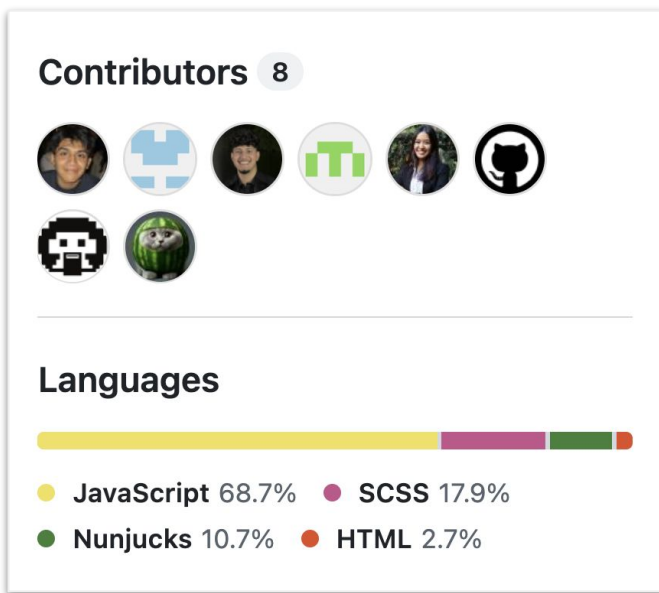
Statistics provided from Lighthouse on Google Chrome developer



OUR IMPACT

All rotation goals accomplished!

1. Fully Open Source codebase
2. Mobile support
3. Vendor-neutrality
4. Automatic dashboard updates
5. Accessibility upgrades
6. **The website is live**
and repo is open for contributions!



REFLECTIONS & LESSONS LEARNED

“Good engineering starts with understanding the problem, not just writing code.” - Haseeb

“Accessibility and usability matter!” - Zion

“The best code is built with the next team in mind, so it is easy to update later” - Julio

“It’s a Civic Technologist’s responsibility to design modern tools for the public” - Kiran

“In civic tech sometimes (not always) it’s better to implement and then ask questions!” - Isaac

THANK YOU!



OSPPO Team

OSPPO Lead

Open Source Software Engineer

Open Source Software Engineer

Open Source Software Engineer

Designer

Remy DeCausemaker

Natalia Luzuriaga

Dinne Kopelevich

Sachin Panayil

Louisa Sholar

OEDA Team Robert Shaffer, Kevin Hodges, Ha Abams, Maria Diacogiannis,

DSAC Team Ange Davis, Christina Schildroth, Andrea Fletcher, Mark DeZalia,
Patrick Trulock, Ming Ligh, Aaron Chafetz

Reveal Team Shen Peng, Pooja Vanju, Nisha Galione

Cassie Rubio + CIF Team

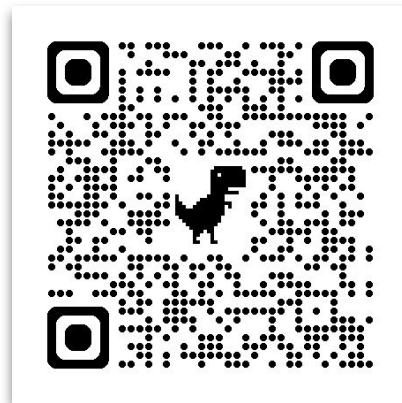
THANK YOU!



Repo



Demo



Deck

Repo: https://github.com/DSACMS/medicare_monthly_enrollment_dashboard

Demo: https://dsacms.github.io/medicare_monthly_enrollment_dashboard

Deck: <https://github.com/DSACMS/decks/blob/main/cif26.pdf>